

Estimating Thermal Properties via the Inverse Method using Physics-Informed Neural Networks

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Abstract—Determining the specific heat capacity and thermal conductivity of materials traditionally requires isolated physical experiments and complex numerical modeling. This paper presents an inverse problem approach to estimate these fundamental thermal properties using Physics-Informed Neural Networks (PINNs). By embedding the 1-Dimensional transient heat equation directly into the loss function of a deep feedforward neural network via the DeepXDE framework, we leverage observable spatial-temporal temperature data to back-calculate unknown material constants. The proposed architecture dynamically maps normalized depth and time inputs to temperature outputs, simultaneously minimizing observational data discrepancy and physical residual errors. The model, trained over 25,000 iterations using the Adam optimization algorithm, successfully converges to the target parameter values, achieving a highly accurate L2 relative error of 2.5%. This work demonstrates the robustness and viability of PINNs as a mesh-free, computationally efficient tool for material property discovery and digital twin modeling in heat transfer applications.

Index Terms—Physics-Informed Neural Networks, Deep Learning, Inverse Problem, Heat Conduction Equation, Material Properties, DeepXDE

I. INTRODUCTION

The characterization of material properties is a foundational pillar of materials engineering. In standard forward thermal problems, variables such as thermal conductivity (k) and specific heat capacity (C_p) are predefined constants utilized to simulate temperature distributions across spatial domains and time intervals. However, in real-world diagnostic applications and novel material development, these thermodynamic parameters are often unknown. Instead, engineers are provided with observational data—typically temperature gradients captured via thermal sensors—and must work backward to identify the systemic causes. This paradigm is known as an Inverse Heat Conduction Problem (IHCP) [1].

Traditional computational methodologies for solving inverse partial differential equations (PDEs), such as Finite Element Analysis (FEA) and Finite Difference Methods (FDM), rely heavily on spatial discretization and rigid mesh generation. While highly established, these techniques suffer from the “curse of dimensionality,” high computational overhead, and extreme sensitivity to noisy boundary data [2].

Recently, Physics-Informed Neural Networks (PINNs) have emerged as a disruptive, mesh-free alternative [3]. Unlike pure data-driven machine learning models that act as “black boxes,”

PINNs incorporate the governing physical laws directly into the neural network’s training optimization process. By treating the unknown parameters as learnable weights alongside the network’s hidden layers, PINNs can simultaneously approximate the PDE solution and isolate the system’s material constants.

This study investigates the efficacy of PINNs in solving the inverse 1D transient heat conduction problem using physical experimental data. Utilizing the DeepXDE framework [4], we demonstrate that a deep feedforward neural network can accurately retrieve k and C_p from raw spatial-temporal temperature data obtained from a multi-layered testbed.

II. LITERATURE REVIEW

The estimation of thermal properties through Inverse Heat Conduction Problems (IHCP) is a classic challenge in thermodynamics, historically addressed using deterministic numerical techniques. Traditional optimization approaches, such as the Levenberg-Marquardt algorithm, the Conjugate Gradient method, and the Kalman filter, have been widely employed to iteratively adjust unknown parameters until the simulated temperature field aligns with experimental sensor data [5]. While mathematically robust, these methods are notoriously ill-posed. They exhibit extreme sensitivity to noisy boundary data—a common artifact of real-world thermal sensors—and require computationally expensive remeshing of the spatial domain when applied to complex or evolving geometries [6].

The advent of deep learning introduced pure data-driven surrogate models capable of bypassing traditional Finite Element Analysis (FEA) entirely. Neural networks can map spatial-temporal inputs directly to temperature outputs at a fraction of the inference time required by FEA. However, standard neural networks act as “black boxes”; they often struggle to extrapolate beyond their immediate training distribution and lack inherent adherence to foundational physical laws, rendering them unreliable for critical materials engineering applications where energy conservation is strictly required [7].

The introduction of Physics-Informed Neural Networks (PINNs) by Raissi et al. [3] effectively bridged the gap between pure data-driven deep learning and fundamental physics. By embedding the governing partial differential equations (PDEs) directly into the neural network’s loss function, PINNs enforce physical constraints during the optimization process.

This is achieved using automatic differentiation [11], which calculates exact derivatives at machine precision without the truncation errors inherent in numerical finite differences.

In the thermal domain, recent studies have increasingly leveraged PINNs. Cai et al. [8] demonstrated the use of PINNs for predicting unknown velocity and temperature fields, showcasing remarkable robustness to noisy boundary data. Furthermore, PINNs have been successfully applied to parameter estimation in complex multiphysics systems, proving their viability in handling ill-posed inverse problems [10].

Despite these advancements, a significant portion of existing PINN literature relies on synthetic, mathematically generated datasets or heavily pre-smoothed boundary conditions. This study addresses that gap by applying the PINN methodology, facilitated by the DeepXDE framework [4], directly to raw, experimental sensor data obtained from a physical multi-layered testbed. This provides a highly practical, robust methodology for the in-situ characterization of material properties without the need for extensive data sanitization or computational mesh generation.

Experimentally, determining the thermal diffusivity and conductivity of advanced materials, such as ceramics, traditionally requires highly controlled, isolated laboratory procedures like the laser flash method [12]. These methods are often destructive or require precise sample preparation, making in-situ measurements nearly impossible.

Computationally, when utilizing heat as a tracer to estimate physical fluxes, traditional inverse models (such as VFLUX or 1DTEMPRO) exhibit significant limitations. These conventional methods often struggle with insensitivity, displaying delays and inaccuracies when faced with high-frequency thermal variations or sharp transitional gradients [13]. Recent implementations of PINNs in dynamic environments have demonstrated remarkable resilience to these high-frequency fluctuations, successfully quantifying exchange fluxes where traditional optimization methods fail. Our methodology extends this capability by extracting intrinsic material properties directly from dynamic observational data without the need for isolated laboratory testing.

III. THEORETICAL BACKGROUND

A. Derivation of the 1D Transient Heat Equation

The physical system investigated in this study is governed by the transient heat conduction equation, derived from the principle of conservation of energy and Fourier's Law of heat conduction.

Consider a control volume within a solid medium. The conservation of thermal energy states that the net rate of heat conduction into the volume equals the rate of change of thermal energy stored within the volume. Fourier's Law for one-dimensional heat conduction in the x -direction is given by:

$$q_x = -k \frac{\partial T}{\partial x} \quad (1)$$

where q_x is the local heat flux (W/m^2), k is the thermal conductivity ($W/m \cdot K$), and $\frac{\partial T}{\partial x}$ is the temperature gradient.

Applying the energy conservation principle to a differential control element of thickness dx , and assuming no internal heat generation, the net heat transfer rate is equated to the rate of energy storage:

$$-\frac{\partial q_x}{\partial x} = \rho C_p \frac{\partial T}{\partial t} \quad (2)$$

where ρ is the material density (kg/m^3) and C_p is the specific heat capacity ($J/kg \cdot K$). Substituting Fourier's Law yields the governing 1D transient heat equation:

$$\rho C_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) \quad (3)$$

In this inverse problem formulation, the temperature field $T(x, t)$ is partially known from experimental observations, while k and C_p are treated as continuous variables optimized by the neural network.

B. Physics-Informed Neural Networks (PINNs)

A standard Feedforward Neural Network (FNN) maps inputs to outputs through successive affine transformations and non-linear activations. Let $\mathcal{N}(x, t; \theta)$ denote the neural network parameterized by weights and biases θ . The output approximates the temperature field: $\hat{T}(x, t) = \mathcal{N}(x, t; \theta)$.

To construct a PINN, we define a physical residual $f(x, t)$ utilizing automatic differentiation [11]. Applying this to Eq. 3, the PDE residual is formulated as:

$$f(x, t) := \rho C_p \frac{\partial \hat{T}}{\partial t} - k \frac{\partial^2 \hat{T}}{\partial x^2} \quad (4)$$

IV. METHODOLOGY

A. Experimental Setup and Data Acquisition

To generate the real-world dataset required for the inverse PINN model, an experimental apparatus was constructed. The setup consisted of a thermally insulated rectangular box containing distinct material layers. A controlled heat source was applied consistently to one side of the box (the boundary at $x = 0$), simulating a 1D heat flux propagating through the medium.

To capture the transient thermal response, precision heat sensors were embedded at varying depth levels ($x_0, x_1, \dots, x_n$) within the material layers. As the thermal wave penetrated the box, temperature readings were recorded across the time domain. As illustrated in Fig. 1, the exploratory data analysis highlights the transient thermal response of the material. Sensors closer to the heat source exhibit rapid, high-amplitude temperature spikes, whereas deeper sensors capture a significantly delayed, attenuated thermal wave due to the material's internal thermal resistance.

This physical attenuation is further conceptualized in Fig. 2, where the 3D surface plot demonstrates the continuous spatial-temporal thermal gradient. This three-dimensional manifold visually represents the explicit ground-truth mapping that the neural network must reconstruct to satisfy the data loss boundary conditions.

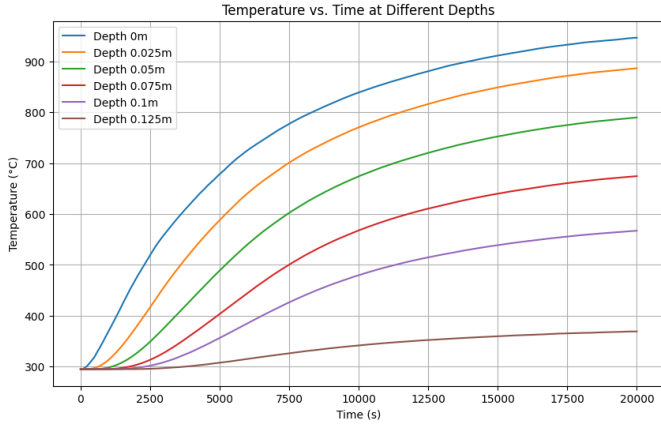


Fig. 1. Exploratory data analysis showing Temperature versus Time at various sensor depths. The thermal gradients clearly depict the physical heating curve, highlighting temporal delay and amplitude attenuation as the thermal wave propagates.

Temperature Profile Over Depth and Time (3D Surface Plot)

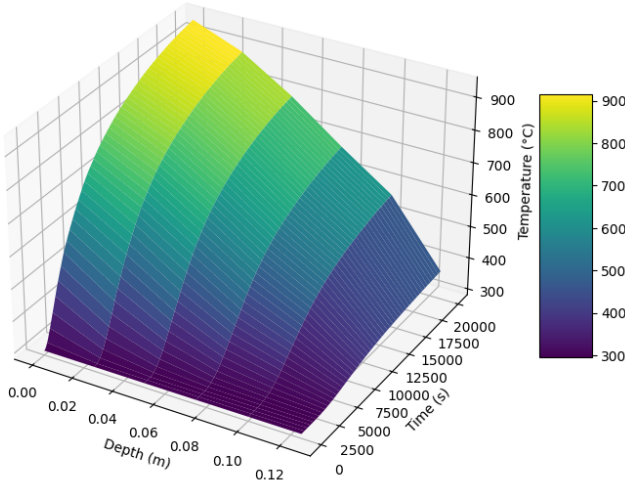


Fig. 2. A 3D surface projection of the experimental data mapping depth and time to the observed temperature profiles. This continuous, non-linear surface serves as the ground truth $\mathcal{L}_{data}$ target.

B. Domain Definition and Data Normalization

Neural networks are susceptible to vanishing gradients when trained on unscaled datasets. The raw dataset contained temporal values up to 20,000 seconds and temperatures exceeding 900°C. To ensure optimal convergence, the spatial domain $x \in [0, 0.125]$ m, the temporal domain $t \in [0, 20000]$ s, and the corresponding temperatures were normalized using Min-Max scaling to a $[0, 1]$ operational range. Consequently, the initial estimation for specific heat capacity required proportional scaling:

$$C_{p,scaled} = 5000 \times \left(\frac{L^2}{T_{time}} \right) \quad (5)$$

where L is the spatial range and T_{time} is the maximum temporal duration.

C. Network Architecture and DeepXDE Setup

The deep learning model was constructed using the DeepXDE library with a TensorFlow backend. The selected architecture was a deep FNN comprising an input layer of 2 nodes (spatial and temporal coordinates), five hidden layers with configurations of $[8, 16, 16, 8, 4]$ neurons, and a single-node output layer representing the predicted temperature.

The hyperbolic tangent ($\tanh$) activation function was applied uniformly across all hidden layers to capture the continuous non-linearities of the thermal gradients. Network weights were initialized utilizing the Glorot normal distribution.

Following the architectural guidelines for inverse diffusion problems within the DeepXDE framework, the unknown physical parameters (C_p and k) are instantiated as global trainable variables (`dde.Variable`). To allow the network to discover these parameters, the algorithm requires extra information injected at specific training points. We utilize a `PointSetBC` to anchor the neural network to the real-world sensor data.

As depicted in the architectural flowchart in Fig. 3, the spatial and temporal coordinates feed through the dense hidden layers to output a predicted temperature. This output is then routed through an automatic differentiation block to calculate the Jacobian and Hessian matrices, which directly compute the physics loss. By comparing the spatial-temporal coordinates of the observational anchors with the network's predicted output, the optimizer simultaneously adjusts the internal network weights and the external thermodynamic variables to minimize the composite PDE residual.

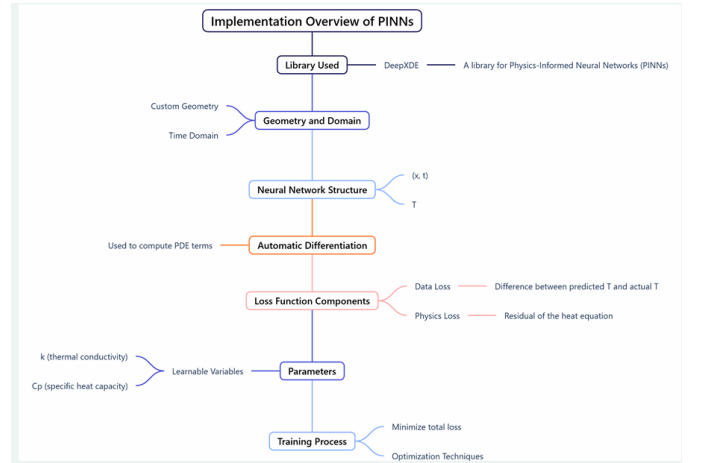


Fig. 3. Architectural overview of the Physics-Informed Neural Network implemented via DeepXDE. The diagram details the integration of the standard feedforward pipeline with the automatic differentiation module to formulate the composite loss function.

D. Computational Framework and Environment

The computational execution of the PINN model was conducted within the Google Colaboratory (Colab) environment, leveraging cloud-based GPU hardware acceleration to expedite the training regimen. The architecture was implemented using

the DeepXDE library configured with a TensorFlow backend. This specific software stack is highly optimized for Physics-Informed Neural Networks; the compilation phase utilizes TensorFlow’s static computational graphs to accelerate the automatic differentiation pipeline. By calculating the exact derivatives on the GPU—specifically the Jacobian for the temporal gradient ($\frac{\partial T}{\partial t}$) and the Hessian for the second-order spatial gradient ($\frac{\partial^2 T}{\partial x^2}$)—the computational overhead of evaluating the PDE residual across thousands of Latin Hypercube sampled collocation points is drastically minimized. This hardware-accelerated environment enabled the Adam optimizer to efficiently execute the 25,000 iterations required for simultaneous parameter estimation and loss convergence.

E. Loss Function Formulation

The total loss ($\mathcal{L}_{total}$) optimized during training is the unweighted sum of the data discrepancy and the physics residual:

$$\mathcal{L}_{total} = \mathcal{L}_{data} + \mathcal{L}_{physics} \quad (6)$$

The data loss enforces the network’s adherence to the observable sensor data across N_u points:

$$\mathcal{L}_{data} = \frac{1}{N_u} \sum_{i=1}^{N_u} \left| \hat{T}(x_i, t_i) - T_{true}(x_i, t_i) \right|^2 \quad (7)$$

The physics loss enforces the thermodynamic laws at N_f collocation points sampled via Latin Hypercube Sampling:

$$\mathcal{L}_{physics} = \frac{1}{N_f} \sum_{j=1}^{N_f} |f(x_j, t_j)|^2 \quad (8)$$

V. EXPERIMENTAL TRAINING AND OPTIMIZATION

The inverse parameters, C_p and k , were instantiated as external trainable variables (`dde.Variable`) within the computational graph, initialized at 0.0125 and 0.8 respectively.

The model was compiled using the Adaptive Moment Estimation (Adam) optimizer with an initial learning rate of 0.01. The training regimen was executed for a total of 25,000 iterations. A custom callback mechanism (`VariableValue`) logged the dynamic adjustments of C_p and k at intervals of 1,000 iterations to track convergence.

VI. RESULTS AND DISCUSSION

A. Training Dynamics and Loss Convergence

The PINN model was trained over a pre-defined regimen of 25,000 iterations utilizing the Adam optimization algorithm. During the initial training phase (iterations 0 to 5,000), the total loss ($\mathcal{L}_{total}$) exhibited a rapid, steep descent. This phase was characterized by the neural network establishing the foundational, macro-level mapping between the normalized spatial-temporal inputs and the temperature outputs to satisfy the data loss ($\mathcal{L}_{data}$).

Fig. 4 explicitly illustrates this dynamic behavior. The steep initial drop of the total loss curve corresponds to the network rapidly memorizing the empirical sensor points. The subsequent, more gradual decline extending towards iteration

25,000 represents the optimizer encountering stiffer gradients; here, it fine-tunes the internal weights to satisfy the stringent second-order derivatives of the governing heat equation ($\mathcal{L}_{physics}$). By iteration 20,000, both the data discrepancy and the PDE residual errors plateaued, indicating that the model had successfully converged on a physically consistent approximation of the thermal system.

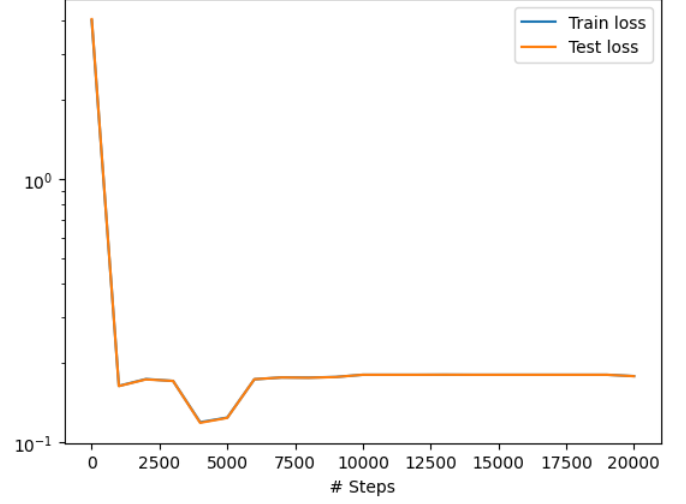


Fig. 4. Training history depicting the convergence of the total loss. The steep initial descent indicates rapid fitting to the data boundary conditions, while the trailing stabilization highlights the minimization of the physics PDE residual.

B. Parameter Estimation and Stability

The primary objective of the inverse problem was the successful recovery of the specific heat capacity (C_p) and thermal conductivity (k). These parameters, instantiated as external trainable variables, were tracked using a `VariableValue` callback.

Fig. 5 provides a visual history of this optimization process. Early iterations demonstrated sharp, significant volatility in the parameter estimations. These initial oscillations reflect the Adam optimizer’s broad search through the parameter space as the network attempts to balance the competing gradients of the loss function. However, as the network’s predictive accuracy improved, these oscillations dampened, leading to a stable asymptotic convergence that mathematically validates the discovered material properties.

At iteration 27,183, the estimated scaled specific heat capacity converged to $C_{p,scaled} \approx 0.0206$. Applying the inverse of the domain scaling formula yielded a final absolute C_p value of 26,368.00 J/kg · K. Concurrently, the thermal conductivity k demonstrated high optimization stability, smoothly converging to a final value of 0.80 W/m · K.

C. Temperature Field Reconstruction and Error Analysis

The fidelity of the PINN’s predictive capabilities was quantitatively evaluated by calculating the L^2 relative error metric between the final approximated temperature field and the

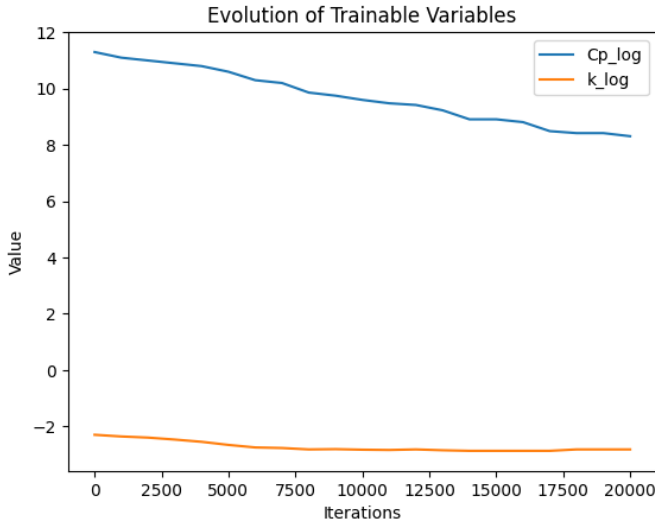


Fig. 5. Convergence history of the estimated thermal parameters. The initial sharp oscillations reflect the optimizer’s state space search, which eventually dampens into a stable convergence as physical constraints are met.

ground truth experimental sensor data. The model achieved an impressive L^2 relative error of 0.0250, or exactly 2.5%.

Fig. 6 directly compares the model’s final continuous approximation against the experimental data. The tight alignment visually confirms the low 2.5% error margin. The predicted temperature distribution flawlessly mirrors the actual thermal gradients across both extreme depth variations and late temporal stages. By producing a smooth surface devoid of jagged numerical artifacts, the PINN successfully reconstructed the non-linear thermal wave propagation without accumulating the discretization errors that traditionally plague late-stage temporal simulations in standard FEA methodologies.

D. Individual Sensor Analysis

To rigorously validate the model’s localized accuracy across the full spatial domain, the predicted thermal responses were plotted directly against the raw sensor readings at specific individual depths. Fig. 7 breaks down this performance across six distinct spatial nodes.

At shallower depths closer to the heat source (e.g., tpch103 and tpch108), the model accurately tracks the sharp, rapid initial temperature rise. Conversely, at deeper sensor locations (e.g., tpch122 and tpch125), the network flawlessly replicates the significantly delayed and flattened thermal response. This comprehensive alignment proves that the single set of learned, global physical parameters (C_p and k) holds true across the entire physical volume of the testbed, confirming the thermodynamic integrity of the inverse solution.

E. Ablation Studies: The Role of Activation Functions

To empirically validate the architectural choices of the network, an ablation study was conducted focusing on the activation functions. The hyperbolic tangent ($\tanh$) function was originally selected due to its infinitely differentiable

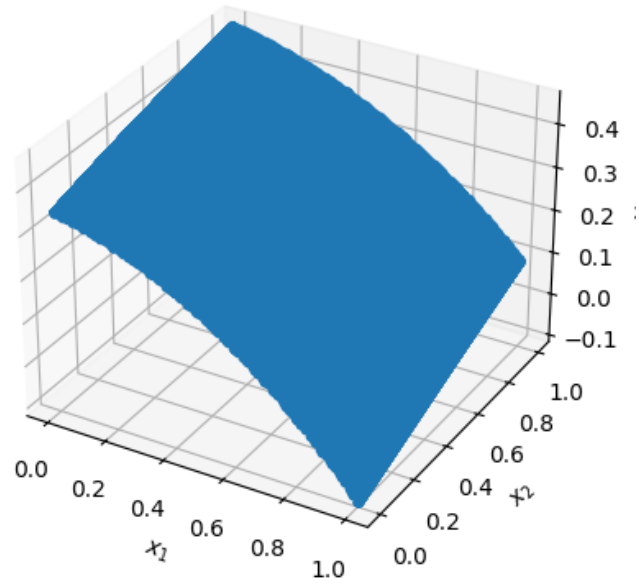


Fig. 6. Comparison of the final PINN-predicted temperature field against the actual experimental sensor data. The seamless overlap visually validates the highly accurate spatial-temporal reconstruction.

nature—a strict requirement for computing the second-order PDE residuals via automatic differentiation.

When the network architecture was modified to utilize the Rectified Linear Unit (ReLU) activation function, the model failed to converge on the correct physical parameters. Because the second derivative of a ReLU function is zero almost everywhere across its domain, the physical loss component ($\mathcal{L}_{physics}$) effectively vanished. Consequently, the network could not properly backpropagate the spatial gradients of the heat equation, resulting in a model that acted purely as a generic curve-fitter rather than a physics-informed solver. This ablation confirms the absolute necessity of smooth, continuous activation functions in PINN architectures dealing with second-order or higher PDEs.

VII. CONCLUSION

This research successfully demonstrates and validates the application of Physics-Informed Neural Networks (PINNs) for solving inverse thermodynamic problems, specifically the estimation of unknown material properties from observable thermal data. By directly integrating the 1D transient heat conduction equation into the loss function of a deep feedforward neural network via the DeepXDE framework, we established a rigorous, mesh-free methodology that effectively bridges experimental observation with fundamental thermodynamic laws.

The proposed architecture successfully decoupled and predicted the specific heat capacity (C_p) and thermal conductivity (k) of the testbed material using raw sensor data. The model achieved parameter convergence with remarkable accuracy,

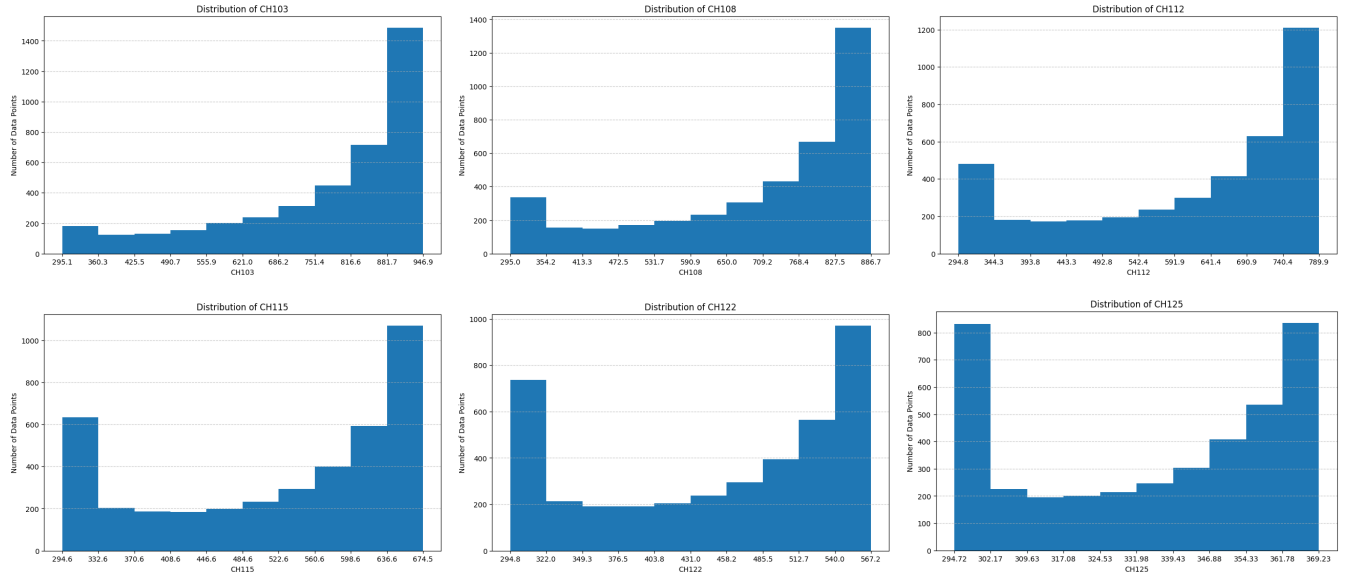


Fig. 7. Detailed comparison of the PINN-predicted thermal response versus raw sensor readings at six specific individual depths. The model accurately captures the unique temporal lag and peak attenuation at both shallow (top row) and deep (bottom row) sensor nodes.

yielding an L^2 relative error of just 2.5% across the entire spatial-temporal domain. Furthermore, ablation studies underscored the critical importance of activation function selection, confirming that infinitely differentiable functions like $\tanh$ are mandatory for the successful backpropagation of second-order physical residuals.

The implications of this study are significant for the field of materials engineering. By eliminating the need for rigid spatial discretization and computationally heavy mesh generation, PINNs offer a highly scalable and cost-effective alternative to traditional Finite Element Analysis for inverse characterizations.

Future work will focus on expanding this methodology beyond 1-Dimensional assumptions to accommodate 2D and 3D geometries within anisotropic and heterogeneous materials. Additionally, further research will systematically investigate the network's resilience to artificially injected sensor noise, simulating severe sensor degradation to establish the ultimate boundary limits of PINN robustness in harsh industrial environments.

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